

Xác định yêu cầu hệ thống

Gv: trần thị minh khoa

30LT + 30TH

2TK + 3TH + GK(thuchanh) + 1CK(tu luan)

diem danh (th) -> 1cot

Thuyet trinh/bcao cao nhom (2->4 mems)

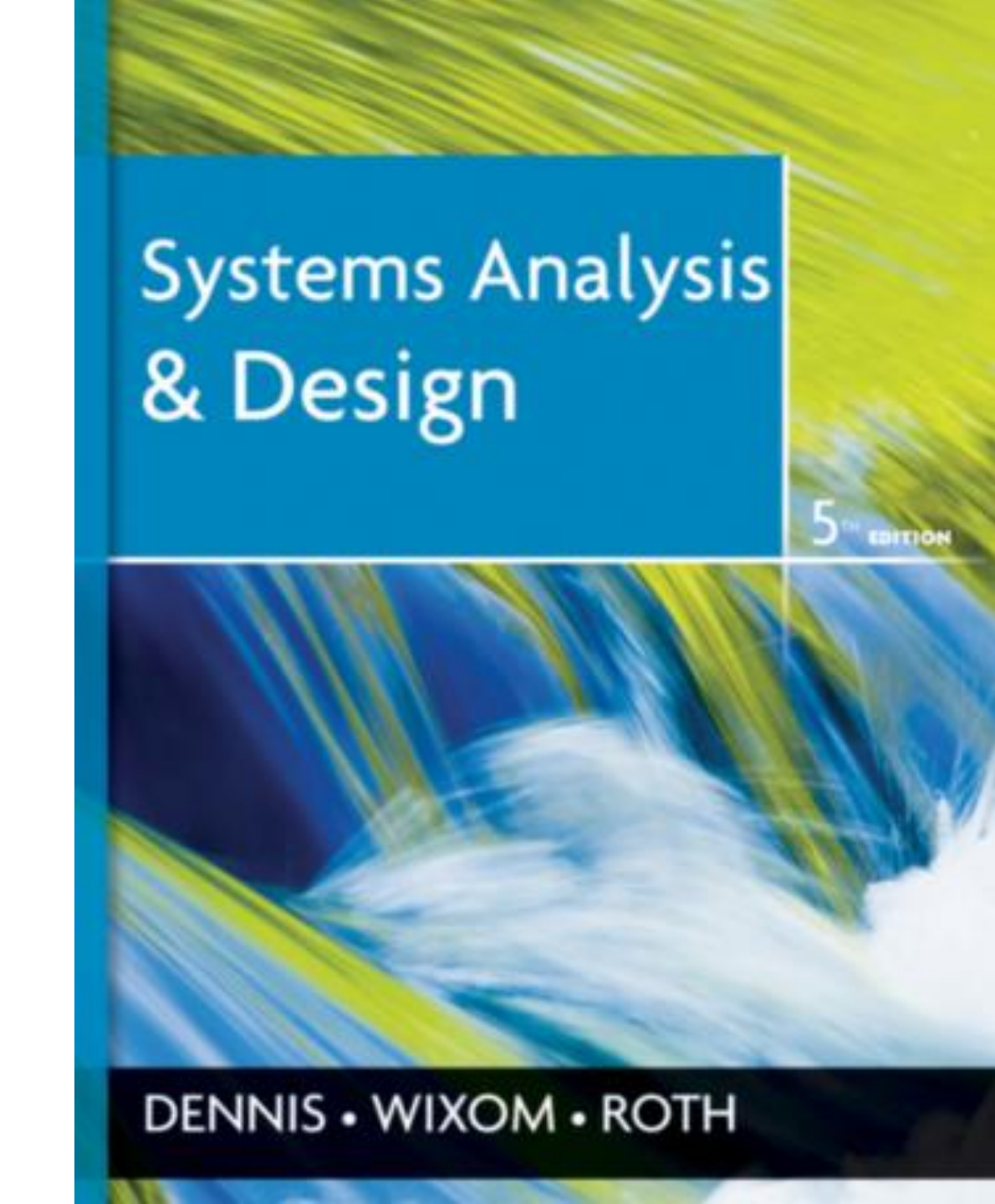
*được sử dụng tài liệu GIẤY viết tay của bản thân*

**Cấm thi:**

*Vắng 4 buổi*

$GK == 0$

$CK \geq 3.0$  (<3.0 ht ko tính điểm tổng kết)



# Systems Analysis & Design

5<sup>th</sup> EDITION

DENNIS • WIXOM • ROTH

# System Analysis & Design

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Presenter: PhD. Tran Thi Minh Khoa

# Agenda

## Planning Phase

1. The Systems Analyst & Information Systems Development
2. Project Selection & Management

## Analysis Phase

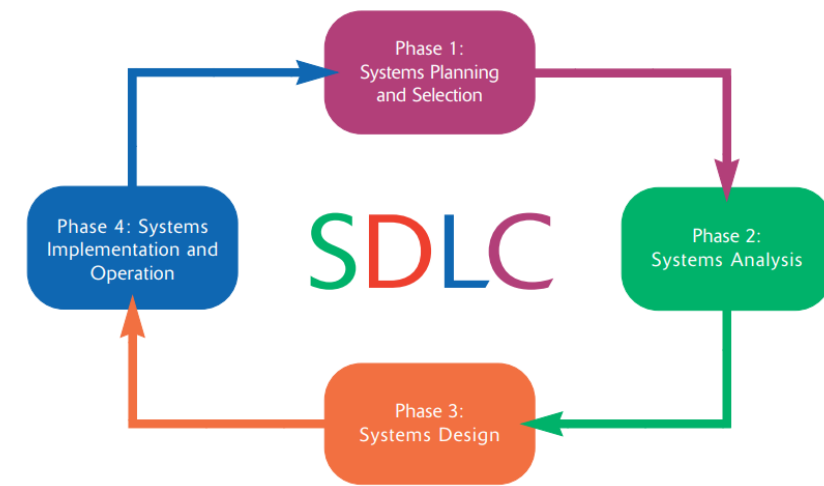
3. Requirement Determination
4. Use Case Analysis
5. Process Modeling
6. Data Modeling

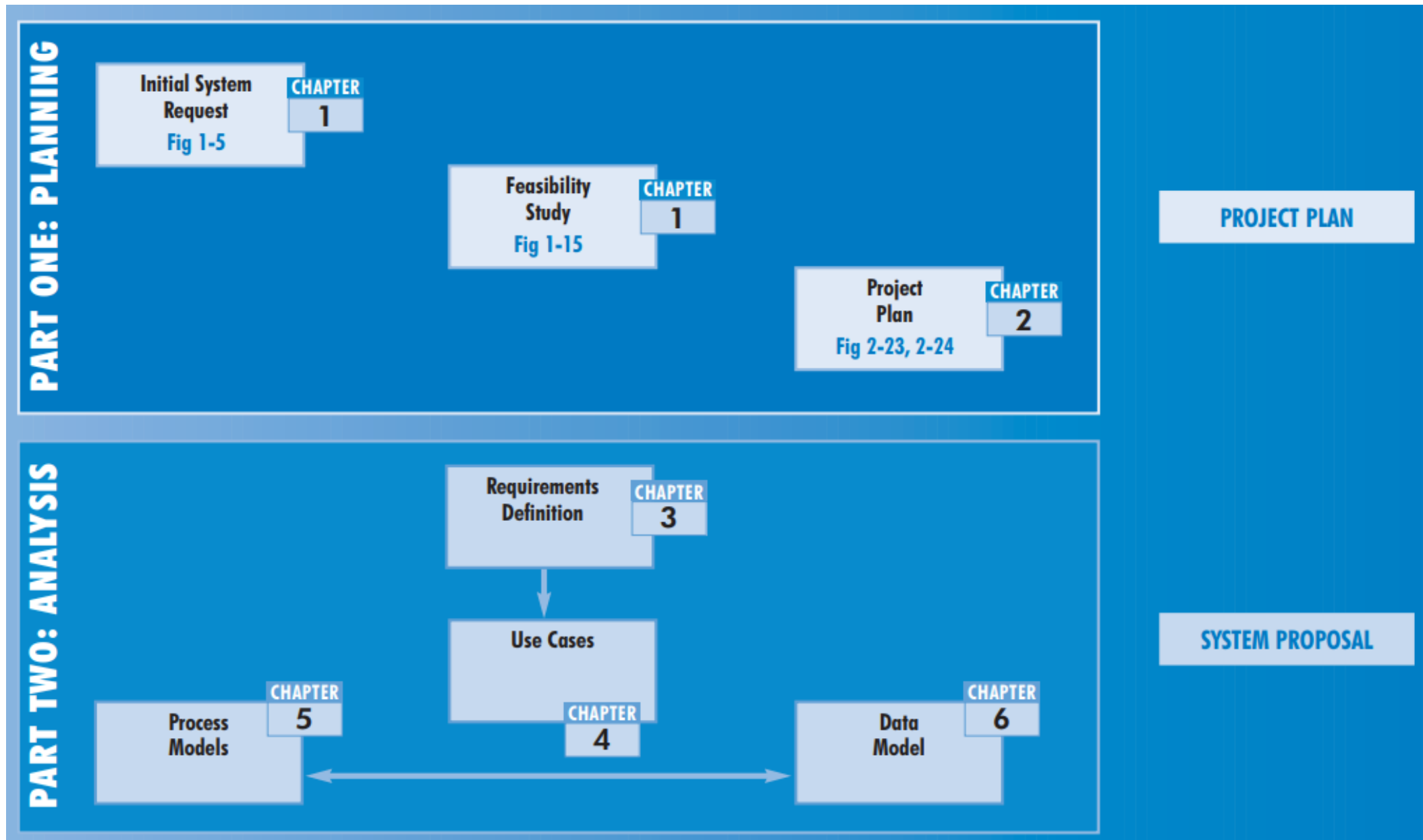
## Design Phase

7. Moving Into Design
8. Architecture Design
9. User Interface Design
10. Program Design
11. Data Storage Design

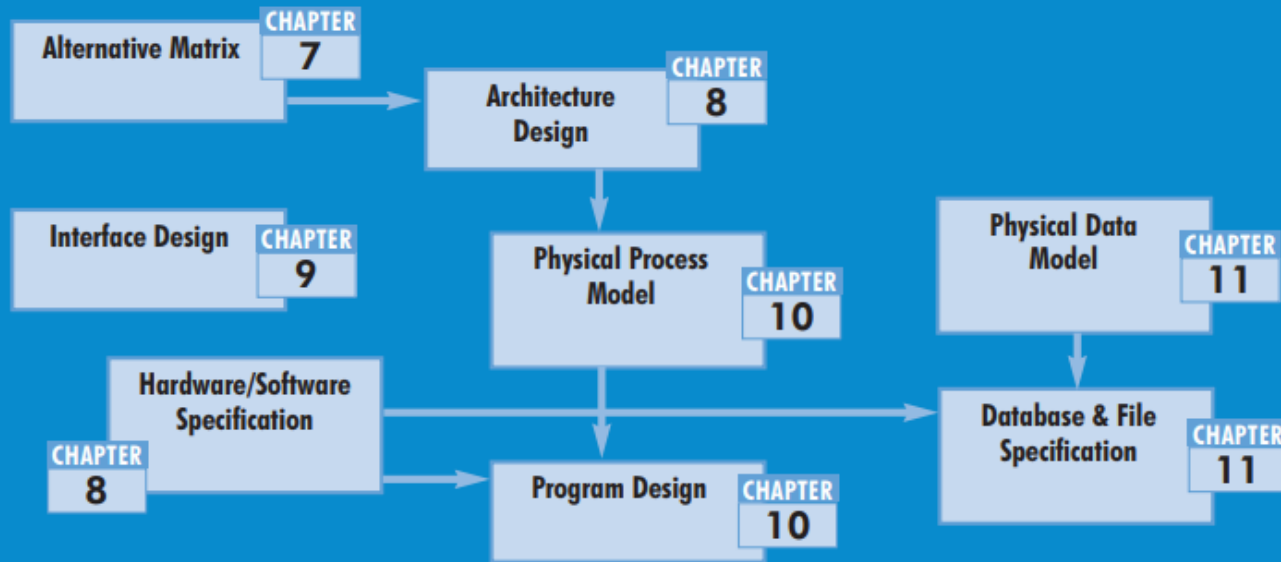
## Implementation Phase

12. Moving Into Implementation
13. Transition To The New System
14. The Movement To Object



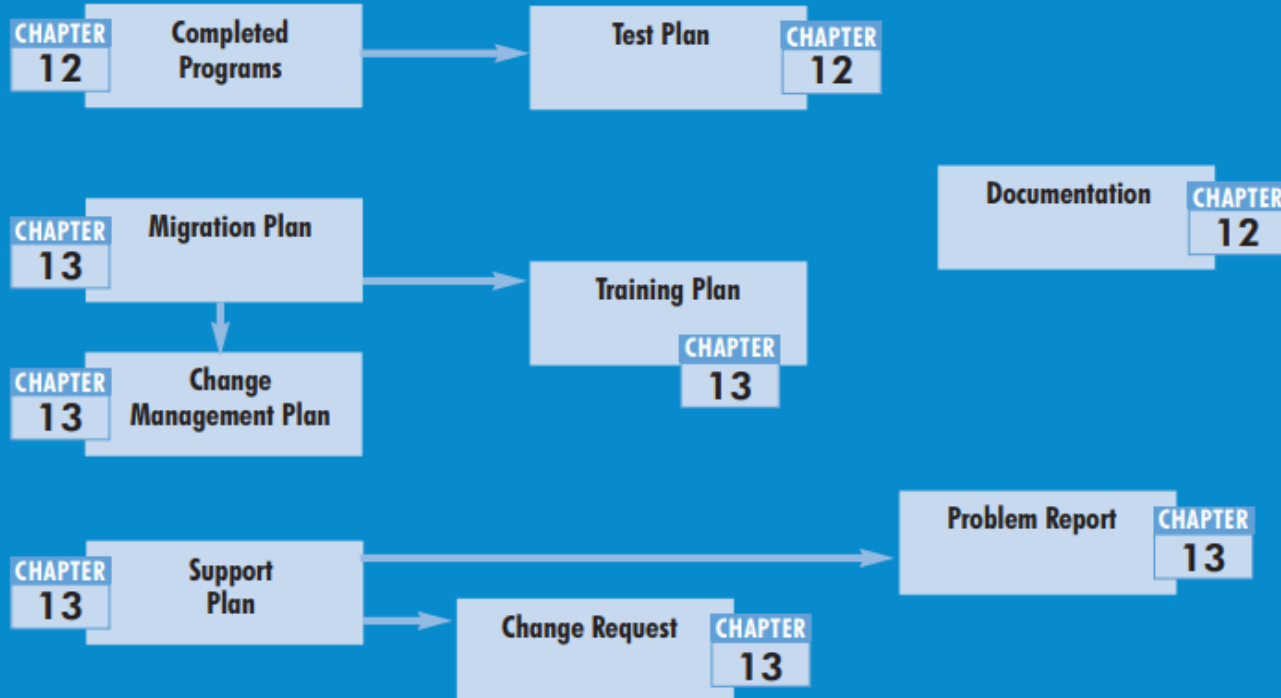


## PART THREE: DESIGN



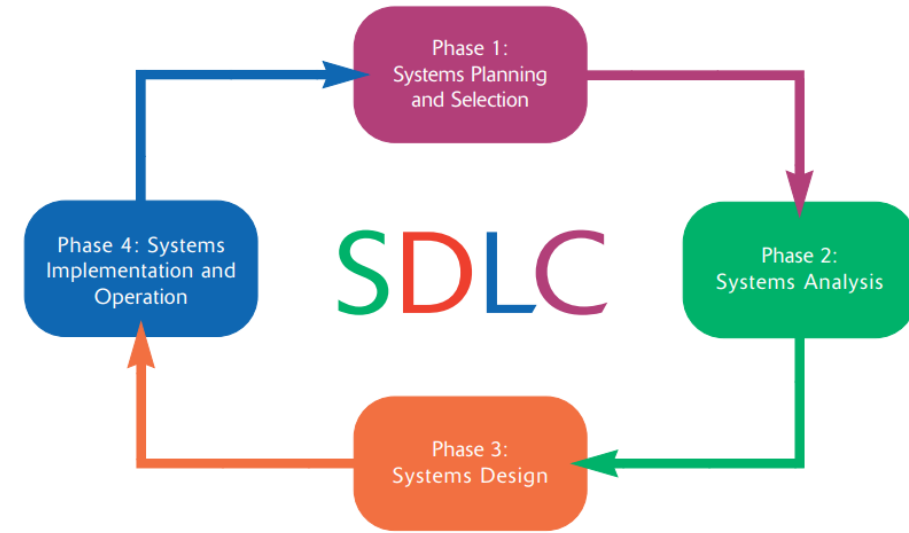
SYSTEM SPECIFICATION

## PART FOUR: IMPLEMENTATION



INSTALLED SYSTEM

# System Development Life Cycle (SDLC)



**The Planning Phase** is the fundamental two-step process of understanding why an information system should be developed and creating a plan for how the project team will develop it.

**The Design phase** decides how the system will operate. This collection of deliverables is the system specification that is handed to the programming team or implementation.

**The Analysis phase** answers the questions of who will use the system, what the system will do, and where and when it will be used.

The final phase in the SDLC is the **Implementation phase**, during which the system is actually built (or purchased, in the case of a packaged software design)

**The Planning Phase**  
is the fundamental two-step process  
of understanding why  
an information system should be developed  
and creating a plan for how  
the project team will develop it.

The deliverables from both steps  
are combined into the project plan,  
which is presented to the project sponsor and  
approval committee at the end of the  
Planning Phase. They decide whether it is  
advisable to proceed with the system  
development project.

# Chapter 1: THE SYSTEMS ANALYST AND INFORMATION SYSTEMS DEVELOPMENT

## OBJECTIVES

- Explain the role played in information systems development by the systems analyst.
- Describe the fundamental systems development life cycle and its four phases.
- Explain how organizations identify IS development projects.
- Explain the importance of linking the information system to business needs.
- Be able to create a system request.
- Describe technical, economic, and organizational feasibility assessment.
- Be able to perform a feasibility analysis.

## CHAPTER OUTLINE

Introduction

The Systems Analyst

*Systems Analyst Skills*

*Systems Analyst Roles*

The Systems Development Life Cycle

*Planning*

*Analysis*

*Design*

*Implementation*

Project Identification and Initiation

*System Request*

*Applying the Concepts at Tune Source*

Feasibility Analysis

*Technical Feasibility*

*Economic Feasibility*

*Organizational Feasibility*

*Applying the Concepts at Tune Source*

Appendix 1A—Detailed Economic

Feasibility Analysis for Tune Source



The *systems development life cycle* (SDLC) is the process of determining how an information system (IS) can support business needs, designing the system, building it, and delivering it to users. If you have taken a programming class or have programmed on your own, this probably sounds pretty simple. In the real world, however, it is not so easy.

In 2010, an estimated \$2.4 trillion was spent by organizations and governments on IT hardware, software, and services worldwide. This spending level was projected to increase by 3.5% in 2011.<sup>1</sup> Unfortunately, a study conducted in 2008 found success is “improbable” in 68% of technology projects.<sup>2</sup> Many of the systems that aren’t totally abandoned are delivered to the users significantly late, cost far more than expected, and have fewer features than originally planned.

A 2009 study attempting to quantify the costs of this failure rate estimated a toll on the global economy of \$6.2 trillion.<sup>3</sup> While this specific outcome has been questioned by some, the point remains that the cost of IT project failures is staggering both in terms of the proportion of projects that fail and the costs of those failures.<sup>4</sup>

Today, both businesses and governments experience embarrassing and costly errors in their information systems. Here is a sample of just a few notable software glitches that occurred in 2010:

- A software error resulted in Toys R Us double billing some shoppers for purchases made on Black Friday.
- Verizon Wireless had to refund \$50 million to customers due to billing system errors.
- Chase banking customers were unable to access their online banking accounts for over 24 hours due to a computer glitch.
- McAfee’s anti-virus software product caused its users’ computers to lock up. McAfee offered affected customers a free 2-year subscription and reimbursement for costs incurred to repair the machines.
- A U.S. Navy drone (unmanned aerial vehicle) reportedly flew into restricted air space near Washington D.C. when operators lost control for about 20 minutes due to a software issue.<sup>5</sup>

# THE SYSTEMS ANALYST

*The systems analyst plays a key role in information systems development projects*

## Systems Analyst Skills

6 major categories:

technical, business, analytical,  
interpersonal, management, ethical.

## Systems Analyst Roles

The *systems analyst* role

The *business analyst* role

The *requirements analyst* role

The *infrastructure analyst* role

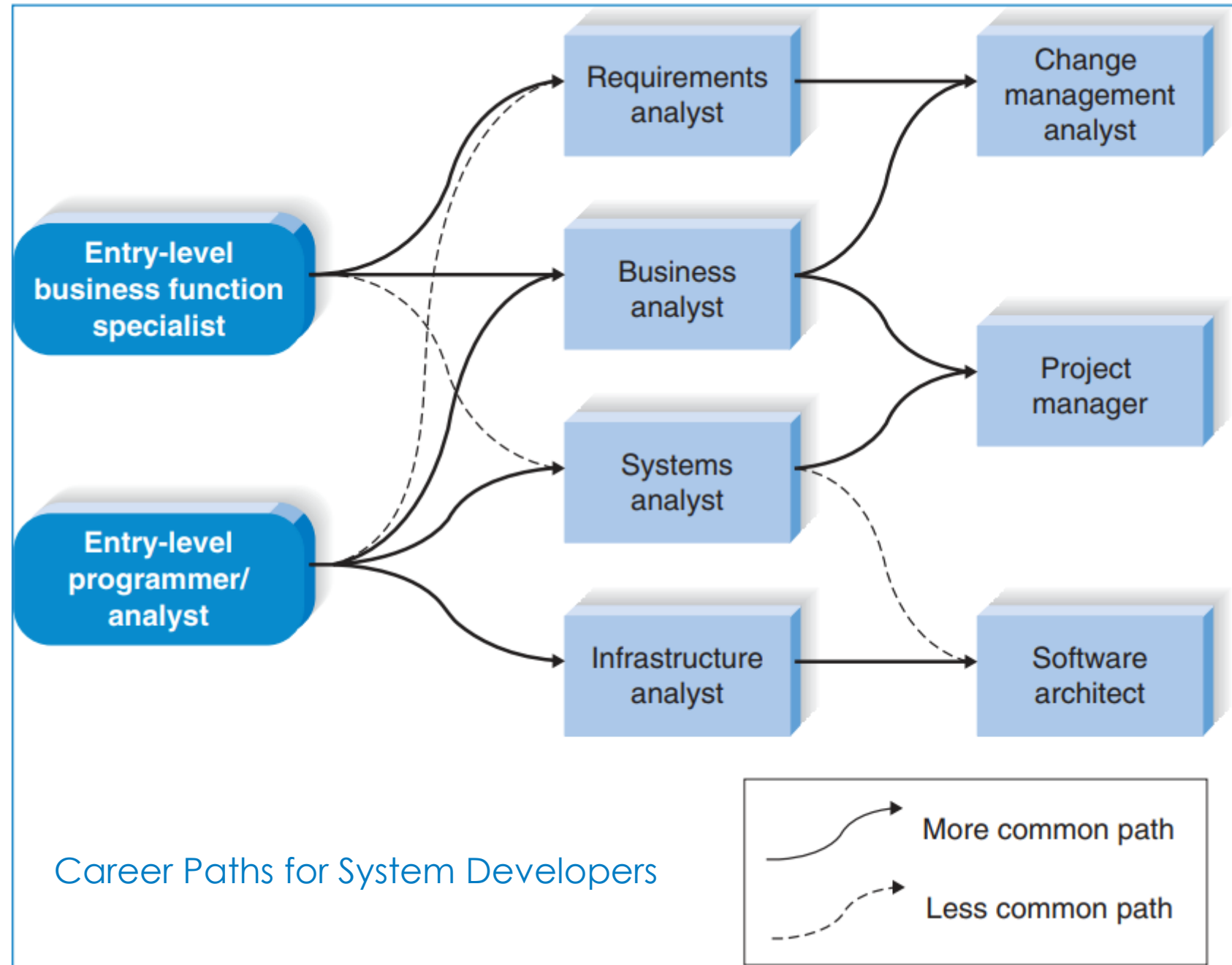
The *change management analyst* role

The *project manager* role

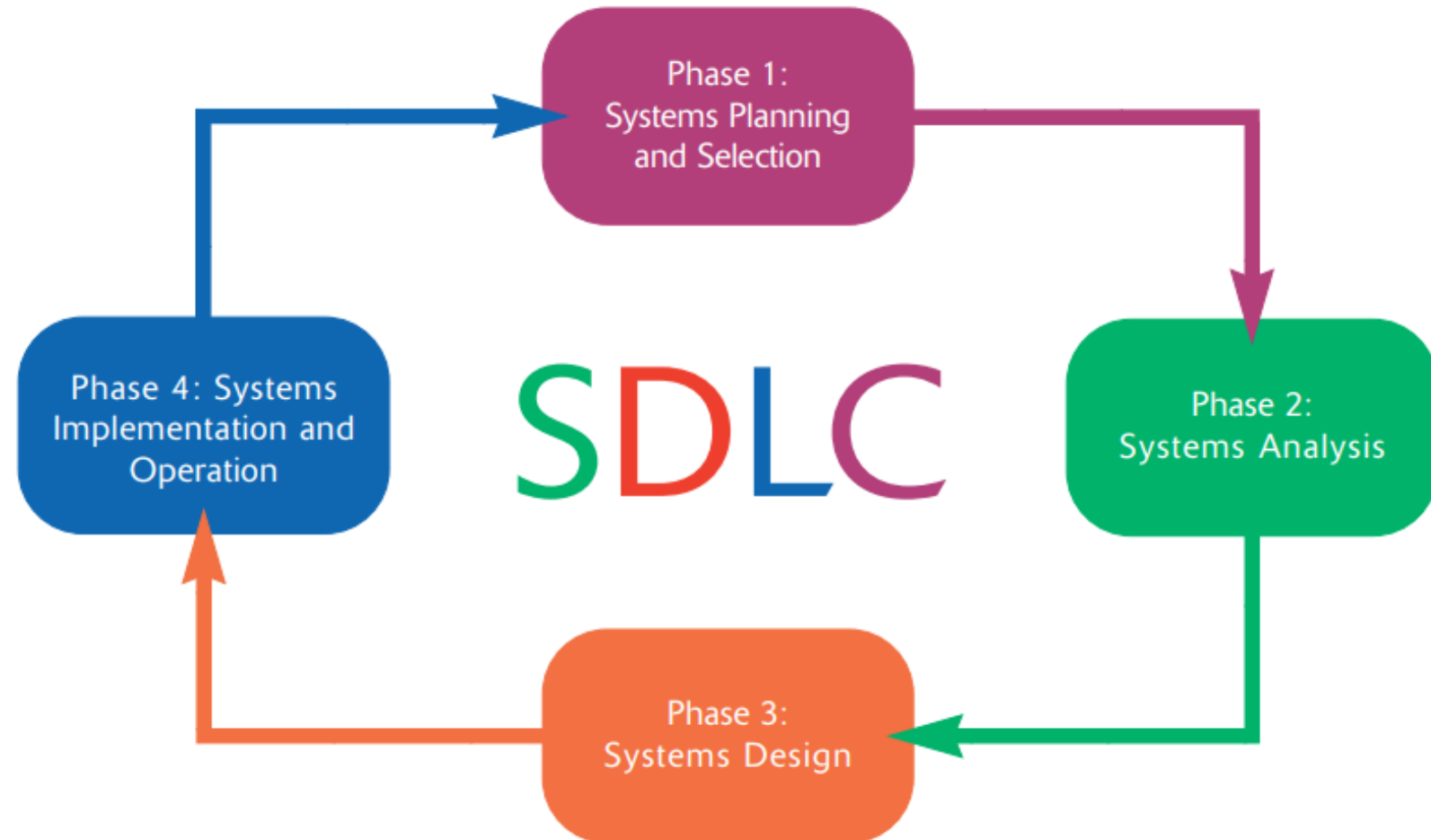
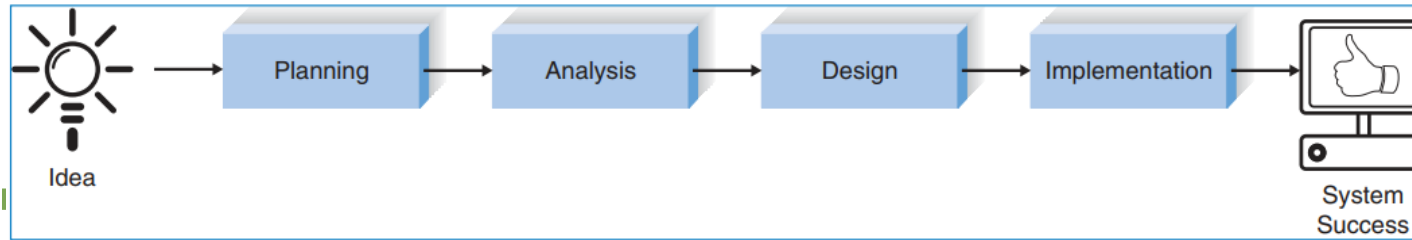
*\*The roles and the names used to describe them may vary from organization to organization.*

“

Some people may enter the field as a more technically-oriented programmer/analyst. Others may enter as a business-oriented functional specialist with an interest in applying IT to solve business problems.



# THE SYSTEMS DEVELOPMENT LIFE CYCLE



# PROJECT IDENTIFICATION AND INITIATION

## Where do project ideas come from?

→ *A project is identified when someone in the organization identifies a business need to build a system.*

→ The *feasibility* (*tính khả thi/thực thi*) analysis examines key aspects of the proposed project:

- The technical feasibility (Can we build it?)
- The economic feasibility (Will it provide business value?)
- The organizational feasibility (If we build it, will it be used?)



- BPM (***Business Process Management***) is a methodology used by organizations to continuously improve end-to-end business processes.
- BPM generally follows a continuous cycle of systematically creating, assessing, and altering business processes.
- Benefits by applying BPM:
  - Enhanced process agility
  - Improved process alignment with industry “best practices”
  - Increase process efficiencies

→ **Project sponsor** is recognized

Including: marketing, accounting, or finance, or also members of the IT area

→ A **system request** is a document that describes the business reasons for building a system and the value that the system is expected to provide

# FEASIBILITY ANALYSIS

## Feasibility analysis

- guides the organization in determining whether to proceed with the project.
- identifies the important risks associated with the project that must be managed if the project is approved

**Include techniques to assess three areas:** technical feasibility, economic feasibility, and organizational feasibility

### **Technical Feasibility: Can We Build It?**

- Familiarity with application: Less familiarity generates more risk.
- Familiarity with technology: Less familiarity generates more risk.
- Project size: Large projects have more risk.
- Compatibility: The harder it is to integrate the system with the company's existing technology, the higher the risk will be.

### **Economic Feasibility: Should We Build It?**

- Development costs
- Annual operating costs
- Annual benefits (cost savings and/or increased revenues)
- Intangible benefits and costs

### **Organizational Feasibility: If We Build It, Will They Come?**

- Project champion(s)
- Senior management
- Users
- Other stakeholders
- Is the project strategically aligned with the business?

Feasibility Analysis Assessment Factors

**Technical Feasibility:** Technical feasibility analysis is, in essence, a *technical risk analysis* that strives to answer the question: “*Can* we build it?”

- Familiarity with the technology
- Project size
- The compatibility of the new system with the technology that already exists in the organization



**Economic Feasibility:** also called a *cost–benefit analysis*, to answer the question “*Should* we build the system?”

- determined by identifying costs and benefits associated with the system,
- assigning values to them,
- calculating future cash flows, and
- measuring the financial worthiness of the project.

Result: the financial opportunities and risks of the project can be understood

Page 25-32

**1. Identify Costs and Benefits**

List the tangible costs and benefits for the project. Include both one-time and recurring costs.

**2. Assign Values to Costs and Benefits**

Work with business users and IT professionals to create numbers for each of the costs and benefits. Even intangibles should be valued if at all possible.

**3. Determine Cash Flow**

Forecast what the costs and benefits will be over a certain period, usually, three to five years. Apply a growth rate to the values, if necessary.

**4. Assess Project's Economic Value**

Evaluate the project's expected returns in comparison to its costs. Use one or more of the following evaluation techniques:

- **Return on Investment (ROI)**

Calculate the rate of return earned on the money invested in the project, using the ROI formula.

- **Break-Even Point (BEP)**

Find the year in which the cumulative project benefits exceed cumulative project costs. Apply the breakeven formula, using figures for that year. This calculation measures how long it will take for the system to produce benefits that cover its costs.

- **Net Present Value (NPV)**

Restate all costs and benefits in today's dollar terms (present value), using an appropriate discount rate. Determine whether the total present value of benefits is greater than or less than the total present value of costs.

Steps to Conduct an Economic Feasibility Analysis

**Organizational Feasibility:** how well the system ultimately will be accepted by its users and incorporated into the ongoing operations of the organization, to answer the question “If we build it, will they come?”

| Role                             |                                                                                                                                                                                                                                                                | To Enhance Organizational Feasibility                                                                                                                                                                                                                                                                                             |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Champion</b>                  | A champion: <ul style="list-style-type: none"> <li>• Initiates the project</li> <li>• Promotes the project</li> <li>• Allocates his or her time to the project</li> <li>• Provides resources</li> </ul>                                                        | <ul style="list-style-type: none"> <li>• Make a presentation about the objectives of the project and the proposed benefits to those executives who will benefit directly from the system.</li> <li>• Create a prototype of the system to demonstrate its potential value.</li> </ul>                                              |
| <b>Organizational Management</b> | Organizational managers: <ul style="list-style-type: none"> <li>• Know about the project</li> <li>• Budget enough money for the project</li> <li>• Encourage users to accept and use the system</li> </ul>                                                     | <ul style="list-style-type: none"> <li>• Make a presentation to management about the objectives of the project and the proposed benefits.</li> <li>• Market the benefits of the system, using memos and organizational newsletters.</li> <li>• Encourage the champion to talk about the project with his or her peers.</li> </ul> |
| <b>System Users</b>              | Users: <ul style="list-style-type: none"> <li>• Make decisions that influence the project</li> <li>• Perform hands-on activities for the project</li> <li>• Ultimately determine whether the project is successful by using or not using the system</li> </ul> | <ul style="list-style-type: none"> <li>• Assign users official roles on the project team.</li> <li>• Assign users specific tasks to perform, with clear deadlines.</li> <li>• Ask for feedback from users regularly (e.g., at weekly meetings).</li> </ul>                                                                        |

## Case study: Applying the Concepts at Tune Source company

### System Request for Tune Source

#### System Request—Digital Music Download Project

**Project Sponsor:** Carly Edwards, Assistant Vice President, Marketing

**Business Need:** This project has been initiated to increase sales by creating the capability of selling digital music downloads to customers through kiosks in our stores, and over the Internet using our website.

**Business Requirements:** Using the Web or in-store kiosks, customers will be able to search for and purchase digital music downloads. The specific functionality that the system should have includes the following:

- Search for music in our digital music archive.
- Listen to music samples.
- Purchase individual downloads at a fixed fee per download.
- Establish a customer subscription account permitting unlimited downloads for a monthly fee.
- Purchase music download gift cards.

**Business Value:** We expect that Tune Source will increase sales by enabling existing customers to purchase specific digital music tracks and by reaching new customers who are interested in our unique archive of rare and hard-to-find music. We expect to gain a new revenue stream from customer subscriptions to our download services. We expect some increase in cross-selling, as customers who have downloaded a track or two of a CD decide to purchase the entire CD in a store or through our website. We also expect a new revenue stream from the sale of music download gift cards.

Conservative estimates of tangible value to the company include the following:

- \$757,500 in sales from individual music downloads
- \$950,000 in sales from customer subscriptions
- \$205,000 in additional in-store or website CD sales
- \$153,000 in sales from music download gift cards

#### Special Issues or Constraints:

- The marketing department views this as a strategic system. The ability to offer digital music downloads is critical in order to remain competitive in our market niche. Our music archive of rare and hard-to-find music is an asset that is currently underutilized.
- Many of our current loyal customers have been requesting this capability, and we need to provide this service or face the loss of these customers' business.
- Because customers have a number of music download options available to them elsewhere, we need to bring this system to market as soon as possible.

**FIGURE 1-6**

Sales Projections for Tune Source  
Digital Music Download Project

| Sales Projections                      |                      |               |                      |            |
|----------------------------------------|----------------------|---------------|----------------------|------------|
|                                        | Individual Downloads | Subscriptions | Cross-Selling of CDs | Gift Cards |
| High-level estimate<br>(prob. = 25%)   | \$900,000            | \$1,100,000   | \$250,000            | \$180,000  |
| Medium-level estimate<br>(prob. = 60%) | 750,000              | 950,000       | 200,000              | 150,000    |
| Low-level estimate<br>(prob. = 15%)    | 550,000              | 700,000       | 150,000              | 120,000    |
| Weighted average<br>expected sales     | \$757,500            | \$950,000     | \$205,000            | \$153,000  |

For example, for individual downloads,

$$\begin{aligned}\text{Expected sales} &= (900,000 * .25) + (750,000 * .60) + (550,000 * .15) \\ &= 225,000 + 450,000 + 82,500 \\ &= 757,500\end{aligned}$$

These projections are summarized in Figure 1-6.

After analyzing the survey results, Carly and her staff were confident that the sales projections and probability estimates were as accurate as they could make them this early in the project. The completed system request is shown in Figure 1-5.



## Feasibility Analysis Executive Summary for Tune Source

*Figure 1-15 presents the executive summary page of the feasibility study: The report itself was about 10 pages long, and it provided additional detail and supporting documentation*

### Digital Music Download Project Executive Summary

Carly Edwards and Jason Wells created the following feasibility analysis for the Tune Source Digital Music Download Project. The System Request is attached, along with the detailed feasibility study. The highlights of the feasibility analysis are as follows:

#### Technical Feasibility

The Digital Music Download system is feasible technically, although there is some risk.

*Tune Source's risk regarding familiarity with music download applications is moderately high.*

- The Marketing Department has little experience with a subscription-based business model.
- The IT department has strong knowledge of the company's existing Web-based CD sales system, but it has not worked with music downloads or customer subscriptions.
- Numerous music download sites exist on the Internet.

*Tune Source's risk regarding familiarity with the technology is moderately low.*

- The IT department has knowledge of the current Web-based order entry system and the databases and Internet technology it uses.
- The IT department has no direct knowledge of the technology required to store and deliver digital music downloads; however, many of the technical issues will be the responsibility of the ISP.
- Consultants are readily available to provide help in this area.

*The project size is considered medium risk.*

- The project team will likely consist of 10 or fewer people.
- Business user involvement will be required.
- The project time frame is somewhat critical, since the system is needed to maintain our competitive position in the market.

*The compatibility with Tune Source's existing technical infrastructure should be good.*

- An Internet infrastructure is already in place at the retail stores and corporate headquarters.
- The ISP should be able to scale its services to accommodate the new Digital Music Download system.

#### Economic Feasibility

A cost-benefit analysis was performed; see attached spreadsheet for details (provided in Appendix 1A). Conservative estimates show that the Digital Music Download system has a good chance of significantly enhancing the company's bottom line.

ROI over 3 years: 280%

NPV over 3 years: \$4,180,431

Breakeven occurs after 0.17 years

*Intangible Costs and Benefits*

Improved customer satisfaction.

Enhanced competitive position through expansion of our brand into the music download market.

#### Organizational Feasibility

From an organizational perspective, this project has low risk. The top executives of the company have a strong interest in the project, and the project champion, Carly Edwards, is a respected and knowledgeable marketing executive.

The users of the system, Internet consumers and in-store kiosk users, are expected to appreciate the entry of Tune Source into the music download arena. Management at the stores may have some concern about lost CD sales; however, since customers have so many other options available for music downloads, this system may prevent our losing those customers to other digital music sources and may provide us with the opportunity to cross-sell those customers from our CD inventory.

#### Additional comments:

- The Marketing Department views this as a strategic system. This system will allow us to leverage our music archive and our well-established market position to establish a presence in the digital music download business. Our customers have been requesting such a capability, and we believe it will be well accepted.
- We should consider hiring a consultant with expertise in similar applications to assist with the project.
- We will need new staff to operate the system and potentially to provide customer service for subscribers and gift-card holders.

# Questions & Exercises



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Lập nhóm (tối đa 4 thành viên)  
Đề xuất và phản biện hệ thống